Identification of Visitors at Tourist Destinations in Yogyakarta Using GPS Data

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***Abstract*—This study utilized GPS data from telecommunications operators to analyze tourist behavior in Yogyakarta, Indonesia. A rule-based identification technique was developed to differentiate domestic tourist activity. The analysis revealed significant visitor pattern variations and highlighted priority destinations such as Gembira Loka and Parangtritis Beach. Preprocessing methods and rule-based classification categorized visitors into tourists (individuals from different sub-districts at tourist locations), working residents (spending 40 hours a week at tourist locations), visitors (spending an average of 3.3 hours at tourist sites), and ordinary residents (not fitting the previous categories). This study offers an essential reference for accurately assessing tourism's economic impact, emphasizing that non-local tourists contribute more significantly to local revenue through spending on accommodation, food, and attractions. The findings assist in planning visits and developing effective tourism policies. The study underscores the importance of precise visitor identification techniques to understand travel patterns and enhance tourism development. The results demonstrate GPS data's potential in tourism research, informing marketing strategies and product development. Future research could integrate additional data sources, such as social media check-ins or transaction records, to improve visitor profiling accuracy. Moreover, studies could investigate the long-term impacts of tourism on local economies and the environment. In Indonesia, working hours regulated by Law No. 13 of 2003 ensure a balance between productivity and leisure, influencing tourist behavior. This study advances the understanding of visitor behavior in Yogyakarta and promotes sustainable tourism development.**

Keywords—GPS Data, Tourist Behavior, Yogyakarta, Rule-Based, Destination Management

I. Introduction

Yogyakarta is one of Indonesia's most popular tourist destinations, attracting millions annually with its natural, cultural, and historical attractions. The increase in the number of tourists reached 2,931.98 percent in June 2022, highlighting various factors that influence fluctuations in the number of tourists[1]. A deep understanding of tourist patterns and behavior is essential for sustainable tourism development. Traveler spatiotemporal data helps in infrastructure planning, destination development, and effective marketing strategies [2][3][4].

Advances in smartphone positioning and location technology have opened up new opportunities to comprehensively track travelers' travel patterns through GPS data [5][6]. However, GPS data from telecom operators still mixes tourists with non-tourists (locals and workers) [7][8]. Therefore, traveler identification techniques are needed to separate this data. Tourism drives economic growth by increasing consumption and investment demand, creating jobs, and stimulating environmental sustainability. However, accurately measuring tourism's economic impact requires distinguishing between tourists and locals, as tourists typically spend more than locals [9].

Tamás Kovalcsik, has developed a method to separate tourists and non-tourists using data from telecommunications operators [10]. They process and analyze passive mobile data with rule-based classification to identify travelers. International traveler identification has been done using citizenship data [11], but identifying domestic travelers from mixed GPS (Global Positioning System) data still requires further research. This study aims to identify and analyze domestic travelers using GPS data to profile users into tourists and residents.

Given the increasing number of visitors to Yogyakarta, we provide information about popular tourist destinations in Yogyakarta, giving visitors references for exciting tourist attractions. The contributions of this paper are:

1) Providing insight into whether visitors at a tourist spot are actual tourists or just residents by profiling visitors into tourists, working residents, visitors, and ordinary residents.
2) Offering insight into priority tourist attractions in Yogyakarta.

The paper is structured as follows. We review related works in Section 2. Next, we explain the methodology in Section 3. We present and discuss the findings in Section 4. Finally, we summarize the findings presented in the paper in Section 5.

II. Related Works

As per the United Nations World Tourism Organization (UNWTO) [13], a tourist is defined as people who are travelling outside their regular tracks and staying for at least one night without receiving any financial remuneration. Travel destinations encompass various purposes such as pleasure, business, health, or other motivations. The UNWTO categorizes travelers as either domestic or international, ased on their travel patterns and characteristics.

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According to Law of the Republic of Indonesia Number 13 of 2003 Article 77 Paragraph 2 [14], employee working hours are regulated into two main patterns: 7 hours per day for six days a week or 8 hours per day for five days a week, for a total of 40 hours per week. This regulation is generally applicable, except for certain types of work with special regulations, ensuring a balance between the productivity of the enterprise and the well-being of workers.

In addition to working hours, there is time spent on leisure activities. Research conducted by Ihwanuddin [15] found that Dalegan Beach attracts many local visitors, but an increase in the number of visitors can reduce the comfort of tourists. This study analyzed the carrying capacity of Dalegan Beach and the characteristics of its visitors using quantitative descriptive methods. From a random sample of 100 tourists, the results showed that this beach can accommodate up to 665 visitors with an average visit duration of 3.3 hours and an optimal number of 266 people per hectare. To prevent overcapacity and maintain comfort, the beach should be temporarily closed when visitors approach the maximum limit. This study emphasizes the importance of visitor management based on carrying capacity to maintain the quality of the tourist experience.

In line with data collection accuracy, Kovalcsik [10] and Saluveer [20] emphasize the importance of removing duplicates and checking data quality in cleaning GPS data to produce reliable datasets. However, GPS data from telecom operators often contains a mix of tourists and non-tourists due to the need for more explicit separation [7][8]. To separate this data, techniques using Mobile Positioning Data (MPD) and Call Detail Records (CDR) have helped determine groups of visitors based on the duration and purpose of the visit [16].

In addition, feature development from GPS data involves creating features for residences, calculating travel distances, and classifying the duration of stays. This study also applied a detection process based on radius and time to identify tourist activities. If a person stays within a radius of 200 meters for 20 minutes, the activity is considered insignificant, based on research by Widyawan et al. [17] and Murti [18].

Meanwhile, Pappalardo [19] explain that the home is the starting point of human activity, and ultimately, wherever a person goes, they will return home. Knowing the house's location is crucial in tourism analysis to understand travel activities. Method like home_location() in the scikit-mobility library predict home locations based on nighttime activity, assuming that people are typically home during that time.

With better visitor identification, destination management can be done more comprehensively. To identify visitors, research by Tamás Kovalcsik [10] and Maarten Vanhoof [12] has developed a rules-based classification method that can be used to process and analyze passive mobile data, such as user activity logs, to distinguish tourists from non-tourists. Gonzalez [11] have also identified international travelers through citizenship data. This study emphasizes the importance of data quality control and technology utilization in achieving sustainable and high-quality destination management [21].

III. METHOD

The algorithm in Fig. 1 profiles travelers using GPS data with a low sampling rate and categorizes the data into two groups: tourists and residents. Tourists are defined as individuals who are in a tourist location and come from different sub-districts, while residents are defined as individuals who come from the same sub-district as the tourist site. The population is divided into three categories: working residents, who spend 40 hours a week at tourist sites; visitors, who spend an average of 3.3 hours at tourist sites; and ordinary residents, who do not fall into the previous two categories.

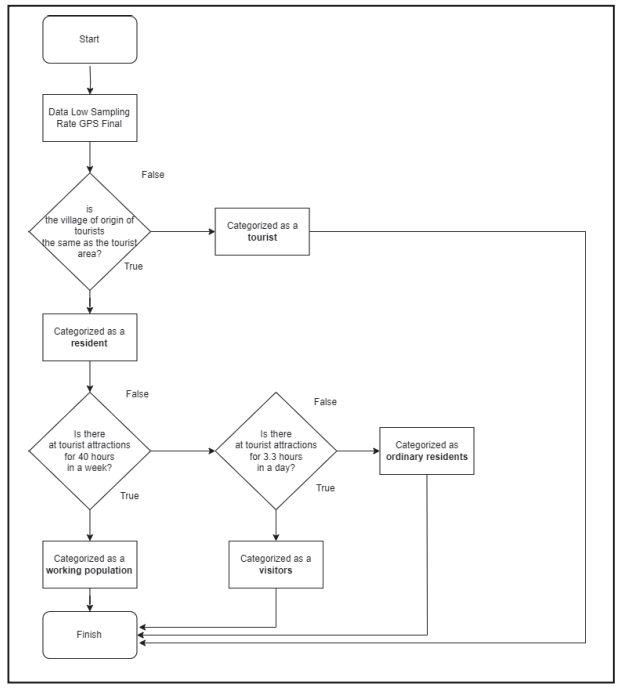

Fig. 1. Flowchart for labeling tourists and residents.

A. Data Collection

Data collection was carried out using a low sampling speed of GPS data from people at tourist attractions. Low sampling rate refers to GPS data collected at longer, irregular intervals. This results in a rarer set of locations, which may only capture a few small details of the user's movements. The advantage of this approach is that it generates smaller volumes of data, making it easier to store, send, and process. This can be particularly useful for applications without critical real-time processing or limited bandwidth and storage [22]. The data used is GPS data from November 2021 to February 2022, a period when Indonesia was still under COVID-19 restrictions, thus limiting the mobility of its population. The GPS data used in this study is sourced from mobile phone users who have activated geolocation services. This creates potential limitations in terms of accuracy and user permissions. Data collection involves processing the initial GPS data, including removing duplicate data, labeling based on time conditions, and changing the data frame field as needed.

B. Data Preparation

This study explores visit patterns in five selected tourist attractions in Yogyakarta based on DIY Tourism Statistics Data. Tourist attractions are selected based on the highest number of visits from each district and municipality in Yogyakarta. The selected tourist attractions are Baron Beach

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in Gunung Kidul, Prambanan Temple in Sleman, Gembira Loka in Yogyakarta, Glagah in Kulon Progo, and Parangtritis in Bantul. We use reverse geocoding techniques to convert geographic coordinates into more detailed location descriptions [23]. This process is carried out by integrating GPS data containing location information with sub-district polygon datasets in Yogyakarta. GPS data frames are enriched with geometric columns based on latitude and longitude columns, which are then converted into GeoDataFrames for geospatial analysis. The administrative region was also converted into a GeoDataFrame, and the two data frames were merged, integrating the location information with the polygons of the administrative area [24].

In preparing geospatial data for analysis, we first identify tourist sites as starting points and then create polygons representing the boundaries of those tourist areas. These polygons are then exported and converted into a shapefile format, combined with the raw data exported and converted into a shapefile format, combined with the raw data, consisting of user ID / MAID, timestamp, latitude, and longitude. The "maid" column is the ID that indicates the user, while the "latitude" and "longitude" columns are used to determine the user's geographic position based on latitude and longitude coordinates. The "timestamp" column is used to determine the time of GPS data collection, thus allowing temporal analysis of the user's movements.

We also implement a home location function to identify the user's home location based on movement data. This process involves collecting location points during the night hours and analyzing them to determine the most frequent locations, which are assumed to be home locations. The next step is to integrate this information into the Yogyakarta Special Region Administrative Region Geometric Dataset to identify the sub-district where the user lives.

Developing features from GPS data includes creating residence features and classifying the duration of stay. The study also applied a detection process based on radius and time threshold to identify tourist activities. If a person stays within a radius of 200 meters for 20 minutes, the activity is considered insignificant, based on research conducted by Widyawan [17] and Murti [18]. Visit data is processed by converting datetime columns to datetime data types, calculating visit duration in hours, and extracting visit start and end times.

The labeling and classification process uses a rule based classification algorithm to distinguish residents from tourists. Data points that show habitual patterns in the exact location at night are considered the location of residents' homes. Data points outside tourist sites are grouped as tourists, allowing us to map and understand the movement dynamics of tourists and locals.

To help visitors establish the priority for tourist visits in Yogyakarta, we have made recommendation to assist travelers in determining the order in which they should visit different tourist attractions in Yogyakarta. We sort data based on unique user identification (UID) and visit date (DateTime) to ensure a consistent order of visits. We eliminate duplicates to avoid data redundancy by removing entries with the same date and travel site name. By taking each UID's first visit, we define each traveler's initial visit. This way, we can calculate the frequency of visits to each tourist location and determine the order of the most prioritized tourist locations based on the number of UIDs. This allows us to provide recommendations that can be very useful for tourists in planning their travel plans, allowing them to choose which attractions to usually visit in advance when they are in Yogyakarta.

C. *Metrics*

To profile tourists, we accommodate several metrics:

1) Home Location Detection and Geometric Data Integration: The home location detection function is used to identify a user's home location from their movement data, assuming they are likely to be at home between 10:00 PM and 4:00 AM. This function analyzes the most frequented points during these hours to determine the approximate location of the house. After determining the latitude and longitude coordinates of each user's home, these coordinates are combined to form the 'house geometry' column, which represents the geospatial position of the house in the geographic information system. This 'house geometry' column is then integrated with the Geometric Dataset of the Yogyakarta Administrative Region, enabling identification of the sub-district (kelurahan) where the user lives and providing a basis for further demographic and geospatial analysis.

2) Temporal and Spatial Thresholds: uses machine learning to analyze traveler mobility with the concept of 'residence location', where a person is considered stationary if they do not move more than 200 meters within 20 minutes. This is an important parameter in mobility analysis [18]. Meanwhile, studying the mobility of the Indonesian population at the beginning of the pandemic and overcoming the inaccuracy of mobile location data by grouping the nearest coordinates as one relevant location, considering that the accuracy of mobile data usually ranges from 200 to 300 meters. Both studies emphasize the importance of temporal and spatial thresholds for accurate analysis of mobility data [17].

3) Visit Duration: The duration of a visit is an essential indicator of tourism activity. start_time comes from a datetime column (for example, "2022-04-03 09:04:13" to "09:04"), marking the beginning of the visit. end_time is calculated from the leaving_datetime column, which records the time when the visit ended.

4) Administrative Region Alignment: To determine if someone is a local or a tourist, we match the location coordinates to their home administrative region. If the location of the house and the tourist site are in the same sub-district, the individual is categorized as a resident. Conversely, if the location is different, they are categorized as tourists. Residents are divided into three categories: working residents (spend 40 hours per week at tourist sites), visitors (spend an average of 3.3 hours per day at tourist sites), and ordinary residents (not included in the previous two categories).

The algorithm in Fig. 2 combines several techniques to identify tourist visit patterns. First, the data is sorted by time, and the first visit to each tourist site is identified. Furthermore, priority tourist locations are determined based on the frequency of choosing each location as the first destination when visiting Yogyakarta.

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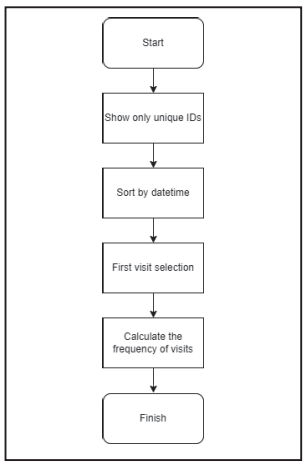

Fig. 2. Flowchart for priority tourism sites.

To evaluate priority tourist sites, we accommodate several metrics:

1) First Visit: Identifies the initial visit by retrieving the first line for each UID.

2) Visit Frequency: Calculate the number of UIDs for each tourist location to determine priority locations.

IV. Results and Discussion

We analyzed tourists at nine tourist sites in Yogyakarta based on low GPS sampling rate data. In profiling site visitors, we use several metrics, including home location, geometric data integration, temporal and spatial thresholds, duration and frequency of visits, and administrative area alignment. We use metrics such as UID, datetime, first visits, and frequency to analyze priority travel locations.

A. Visitor category comparison

The graph in Fig. 3 to Fig. 7 shows the percentage distribution of visitors in the five main tourist attractions in Yogyakarta, such as Prambanan Temple, Gembira Loka, Baron Beach, Parangtritis, and Glagah, based on visitor categories: ordinary residents, tourist workers, visitors, and tourists, from November to February. Prambanan Temple is dominated by ordinary residents every month with a percentage that is always above 60%, but there is a significant increase in tourists in February (13.7%), which can be linked to the Chinese New Year celebration at Prambanan Temple which is open until 17.00 WIB [25]. In addition, in November there is the Prambanan Jazz Virtual Festival [26] and in December-January there is a colossal performance that also increases the number of visits [27]. Gembira Loka also shows the dominance of the ordinary population, but with a fairly high percentage of workers in November (23.6%) and February (33.6%). In December, there are ZOOPEDA activities that may attract more visitors [28]. A study from the University of Warwick shows that a visit to the zoo can increase children's knowledge of science and conservation [29], which also adds to the appeal of Gembira Loka as an educational destination. Baron Beach showed an upward trend in tourists during December and January, reaching 37.6% of tourists in December and 23.1% in January. Parangtritis also showed an increase in tourists in December (19.4%) and January (30.1%). Parangtritis has a special event in November that celebrates the 75th Anniversary of Parangtritis [30] and implements an odd-even system on New Year's Eve to manage tourist crowds [31]. During the Chinese New Year holiday in February, Parangtritis Beach becomes a favorite of tourists, increasing the percentage of visits. Glagah has a unique characteristic with a very high prevalence of ordinary residents in January (94.6%), but it also shows an increase in tourists in December (37.5%) and February (14.3%). However, it should be noted that the data in Glagah is incomplete due to signal constraints in the area, which can affect the accuracy of the recorded distribution of visitors [32]. This explanation shows that each tourist attraction has different visitor characteristics, with interesting variations in terms of the number of workers, locals, and tourists visiting in certain months. This information is very useful for tourist attraction managers to plan more effective operational and marketing strategies, according to the identified visit patterns.

B. Priority Tourism

The interactive map in Fig. 8 depicts the distribution of visitors at various tourist sites in Yogyakarta. The map uses the color and size of the circle to represent the number of visits to each location, with the color scale on the right side of the map showing the range of visits from blue (lowest) to yellow (highest). Gembira Loka and Parangtritis are marked with a large yellow circle, indicating a very high number of visits, reflecting their popularity as the main tourist destinations in Yogyakarta. Other locations such as Prambanan Temple, Baron Beach and Glagah are characterized by smaller, darker circles, indicating a lower number of visits. This map illustrates how Gembira Loka and Parangtritis are the main points of tourist attraction in Yogyakarta, with Prambanan Temple, Baron Beach and Glagah also serving as magnets for tourists visiting Yogyakarta, providing important insights for tourism managers in planning marketing strategies and destination development.

V. Conclusion

The study offers insight into tourism trends and patterns in Yogyakarta, especially in distinguishing between domestic tourists and residents using GPS data. Despite the low sampling rate, the study developed a rule based approach to grouping tourists and tracking tourism activities. The findings show that Gembira Loka and Parangtritis Beach are the leading destinations for first-time visitors to Yogyakarta. Recent advances in GPS and mobile data allow for more comprehensive tracking of tourism patterns, although challenges still need to be addressed in distinguishing tourist data from non-tourist data. This research addresses this challenge by developing preprocessing methods for effective data classification based on specific characteristics. The increasing number of tourists highlights the importance of understanding tourist behavior for sustainable tourism development. Information about priority destinations helps traveler is planning their visit. In conclusion, the study successfully distinguish tourists, regular workers, casual visitors, and residents using GPS data, providing a basis for assessing the economic impact. Tourists from other regions tend to spend more money, directly contributing to the area's income. The results offer valuable insights for tourism development in Yogyakarta, boost the tourism economy, and guide tourists in choosing key destinations.

Future research may explore integrating additional data sources, such as social media check-ins or transaction records, to further improve the accuracy of visitor profiling. In addition, research can examine the long-term impact of

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tourism on the local economy and the environment. Law Number 13 of 2003 in Indonesia regulates working hours to maintain a balance between productivity and worker welfare. This regulation has an impact on the time available for recreational activities, thus affecting the behavior of tourists. Therefore, future research may also investigate how these working-hour regulations affect tourism patterns and visitor numbers.

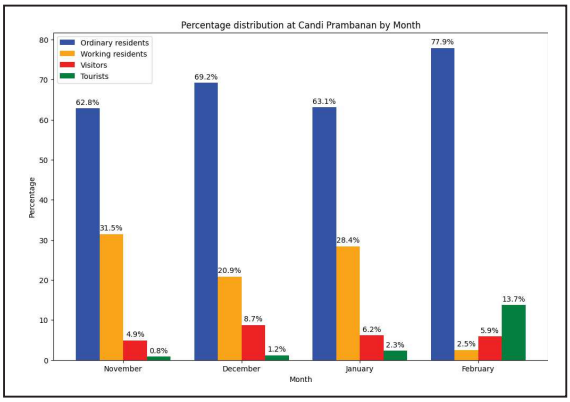

Fig. 3. Visualization of Prambanan Temple.

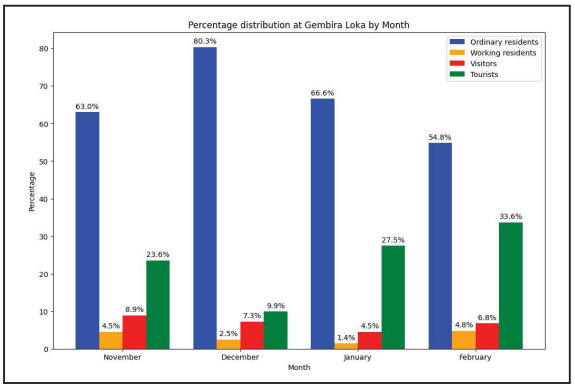

Fig. 4. Visualization of Gembira Loka.

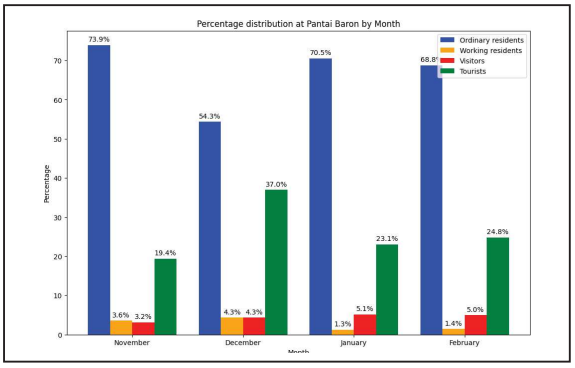

Fig. 5. Visualization of Pantai Baron.

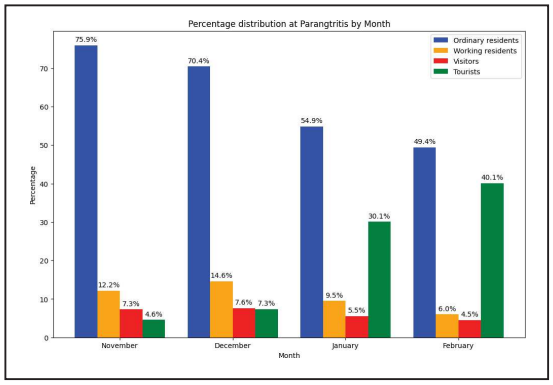

Fig. 6. Visualization of Parangtritis.

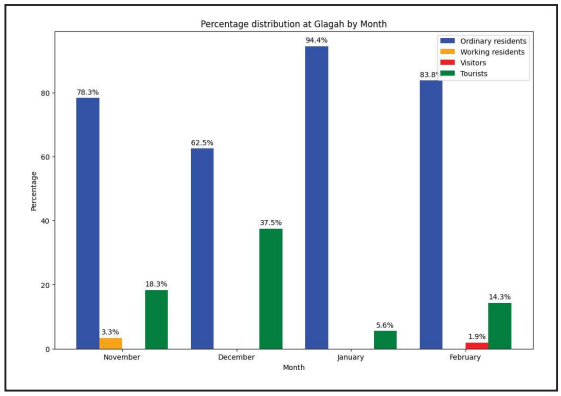

Fig. 7. Visualization of Glagah.

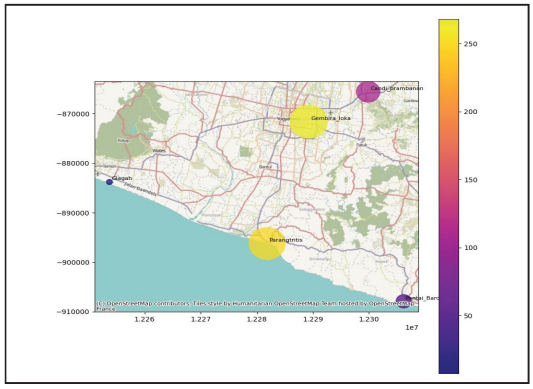

Fig. 8. Visualization of priority tourism.

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